Hints: Linear algebra – 7. Euclidean spaces, orthogonality and the SVD

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Diagonalise $A^TA \succeq 0$ orthogonally: $A^TAv_i = \sigma_i^2 v_i$. For $\sigma_i > 0$ put $u_i = Av_i/\sigma_i$, check that these are orthonormal, and complete them to a basis.

Hint 2. In the bases v_i and u_i, the map sends $\sum x_i v_i$ to $\sum \sigma_i x_i u_i$. So the unit ball goes onto the (possibly degenerate) ellipsoid with semiaxes σ_i. For $Ax = \lambda x$ (complex x), $\|Ax\|^2 = x^*A^TAx$ lies between $\sigma_n^2\|x\|^2$ and $\sigma_1^2\|x\|^2$.

Problem 2

Hint 1. Operator norm: if $\operatorname{rank} B \leqslant k$, then $\ker B$ has dimension $\geqslant n-k$ and meets $\operatorname{span}(v_1, \dots, v_{k+1})$ in some unit vector x. Then $\|(A-B)x\| = \|Ax\| \geqslant \sigma_{k+1}$.

Hint 2. Frobenius norm: use Weyl's inequality $\sigma_{i+j-1}(A) \leqslant \sigma_i(A-B) + \sigma_j(B)$, which follows from the minimax principle. With $j = k+1$ and $\sigma_{k+1}(B) = 0$ you get $\sigma_{i+k}(A) \leqslant \sigma_i(A-B)$. Square and sum.

Hint 3. Truncating the SVD attains both bounds. The singular matrices are exactly those of rank $\leqslant n-1$, so the distance is σ_n.

Problem 3

Hint 1. (IMC 2008.) By the polarisation identity $\langle x-y, x'-y\rangle = \frac{1}{2}(|x-y|^2 + |x'-y|^2 - |x-x'|^2)$, orthogonality is automatic once $|x-y| = d/\sqrt{2}$ for all $x \in S$. So you only need a point y at distance $d/\sqrt{2}$ from every point of S.

Hint 2. Finite $S = \{x_1, \dots, x_m\}$: the centroid c has $|x_i - c|^2 = \frac{m-1}{2m}d^2$. Move from c in a direction orthogonal to the affine span of S, which is possible in infinite dimension.

Hint 3. Infinite S: fix $x_0 \in S$ and a sequence $x_1, x_2, \dots$ in S. Put $u_j = \frac{\sqrt{2}}{d}(x_j - x_0)$, so $|u_j|^2 = 2$ and $\langle u_i, u_j\rangle = 1$. The averages $a_N = \frac{1}{N}\sum_{j\leqslant N} u_j$ form a Cauchy sequence, with $\|a_N - a_M\|^2 = \frac{1}{N} - \frac{1}{M}$ for $N < M$. Their limit f has $|f| = 1$ and $\langle f, u\rangle = 1$ for every $u = \frac{\sqrt{2}}{d}(x - x_0)$, $x \in S \setminus \{x_0\}$. Take $y = x_0 + \frac{d}{\sqrt{2}}f$.

Problem 4

Hint 1. Show $\ker A^* \subseteq \ker A$ by rewriting the equation as $(A^2 - E)A^* = A$. So $\ker A = \ker A^*$, and $W = (\ker A)^\perp$ is invariant under both A and A^*, with A invertible on W.

Hint 2. On W, the equation gives $A^2 - E = AA^{*-1}$, which is invertible. So $A^* = (A^2 - E)^{-1}A$ commutes with A, and A is normal.

Hint 3. For an eigenvalue λ: $\lambda + \bar\lambda = \lambda^2\bar\lambda = \lambda|\lambda|^2$, which forces $\lambda \in \mathbb{R}$. A normal matrix with real eigenvalues is Hermitian.

Problem 5

Hint 1. The answer is $n(n+1)/2$, attained by the symmetric matrices. Write $A = S + K$ (symmetric plus skew). Check that A is t-normal iff $SK = KS$.

Hint 2. Let L be such a space and $Q = L \cap \{\text{skew}\}$. If $A = S + K \in L$ and $K_0 \in Q$, then applying the criterion to $A + K_0$ gives $[S, K_0] = 0$. So the projection of L to the symmetric part lies in $C = \{S \text{ symmetric} : [S, K_0] = 0 \ \forall K_0 \in Q\}$, and $\dim L \leqslant \dim C + \dim Q$.

Hint 3. For $S \in C$, $K_0 \in Q$ and symmetric X: $\operatorname{tr}(S[K_0, X]) = \operatorname{tr}([S, K_0]X) = 0$. So C is orthogonal to $[Q, \mathrm{Sym}]$ for the nondegenerate form $\operatorname{tr}(XY)$ on symmetric matrices. Fix D diagonal with distinct entries. Then $K_0 \mapsto [K_0, D]$ is injective on skew matrices, so $\dim[Q, \mathrm{Sym}] \geqslant \dim Q$.

Problem 6

Hint 1. Transpose $A^T = E - A^2$ and substitute back: $A = E - (E - A^2)^2$, i.e. $A(A - E)(A^2 + A - E) = 0$. In particular A has real eigenvalues.

Hint 2. A^T is a polynomial in A, so A is normal. A real normal matrix with real eigenvalues is symmetric. Hence $A^2 + A = E$, and the eigenvalues are $\frac{-1 \pm \sqrt{5}}{2}$.

Hint 3. Three numbers from $\{\frac{-1 \pm \sqrt{5}}{2}\}$ cannot sum to 0, because $\sqrt{5}$ is irrational. Answer: no.

Problem 7

Hint 1. A is a linear bijection mapping the cube $[-1, 1]^n$ (the unit ball of f) onto itself. So it permutes the extreme points, i.e. the 2^n vertices $\{\pm 1\}^n$.

Hint 2. Some power A^m fixes every vertex, e.g. m equal to the order of this permutation. The vertices span $\mathbb{R}^n$, so $A^m = E$.

Problem 8

Hint 1. Use the canonical form: in a suitable orthonormal basis an orthogonal A with $\det A > 0$ is a direct sum of rotations $R(\theta_j)$ of pairwise orthogonal planes and the identity on the rest (the eigenvalue -1 has even multiplicity and is grouped into rotations by π).

Hint 2. The period of v is the lcm of the orders of the rotations in the planes where v has a non-zero component. For each v_i pick such a plane with rotation order $d_i > 1$, d_i dividing the period of v_i. The periods are pairwise coprime, hence so are the d_i, and the planes are distinct: $n \geqslant 2s$.

Hint 3. Rotations of orders $2, 3, 5, \ldots$ (the first s primes) in s orthogonal planes give vectors with periods $2, 3, 5, \ldots$. Answer: $n = 2s$.

Problem 9

Hint 1. From the SVD, $A = U\Sigma V^T = (UV^T)(V\Sigma V^T)$. Uniqueness of the positive semidefinite square root: any $S \succeq 0$ with $S^2 = P$ commutes with P. So it preserves each eigenspace of P, and on the μ-eigenspace it must be $\sqrt{\mu}\, E$. If A is invertible, $Q = AS^{-1}$.

Hint 2. Define $A^+ = V\Sigma^+ U^T$, where Σ^+ inverts the nonzero σ_i and transposes. The four identities reduce to the diagonal case. AA^+ is the orthogonal projection onto $\operatorname{Im} A$, so $x = A^+ b$ minimises $\|Ax - b\|$. All minimisers form $x + \ker A$, and $A^+ b \in (\ker A)^\perp$ is the shortest.

Problem 10

Hint 1. f is $\mathbb{R}$-linear on the real vector space $M_n(\mathbb{C})$ of dimension $2n^2$, so injective $\iff$ surjective.

Hint 2. Put $Z = X + iY$ with X, Y real. Then $f(Z) = (A+B)X + i(A-B)Y$. So f is injective iff $X \mapsto (A+B)X$ and $Y \mapsto (A-B)Y$ are injective, i.e. iff $A \pm B$ are invertible.